

Indian Institute of Technology, Madras
Department of Applied Mechanics & Biomedical Engineering
Computational Tools: Algorithms, Data Structures and Programs -
ID6105

Assignment set - V

To be submitted by: Thursday 30th October, 2025.

1 Problem 1

Number representation: In class we looked at representing real numbers using different data types on a computer. Let us work out a few examples to understand them,

(a) Convert the following base-2 numbers to base-10: (i) 1011001, (ii) 110.00101 and (iii) 0.01011.

(b) Convert the following base-8 numbers to base -10: (i) 71,563 and (ii) 3.14.

(c) The infinite series $f(n) = \sum_{i=1}^n i^{-4}$, converges on a value of $f(n) = \pi^4/90$ as n approaches infinity. Write a program in single precision to calculate $f(n)$ for $n = 10,000$ by computing the sum from $i = 1$ to 10,000. Then repeat the calculation but in reverse order-that is, from $i = 10,000$ to 1 using increments of -1 . In each case, compute the true percent relative error. Explain the results.

(d) Calculate the random access memory (RAM) in gigabytes necessary to store a multidimensional array that is $200 \times 400 \times 520$. This array is double precision, and each value requires a 64-bit word. Recall that a 64-bit word = 8 bytes and 1 kilobyte = 2^{10} bytes. Assume that the index starts at 1.

(e) What is the maximum size of a multidimensional array $M \times N \times P$ that can be stored on your laptop. Does it tally with your RAM configuration on your laptop? Explain your result.

2 Problem 2

Determine the roots of the simultaneous nonlinear equations using (a) fixed-point iteration and (b) the Newton-Raphson method,

$$y = -x^2 + x + 0.75, \tag{1}$$

$$y + 5xy = x^2. \tag{2}$$

Employ initial guesses of $x = y = 1.2$ and discuss the results.

3 Problem 3

The “divide and average” method, an old-time method for approximating the square root of any positive number a , can be formulated as

$$x = \frac{x + a/x}{2}. \quad (3)$$

Proove that this is equivalent to the Newton-Raphson algorithm.

4 Problem 4

You are designing a tank to hold water as shwn in Fig. 1. The volume of liquid it can hold can be computed as,

$$V = \pi h^2 \frac{(3R - h)}{3}, \quad (4)$$

where V is the volume of the tank (in m^3), h = depth of water in tank (in m), and R = the tank radius (in m). If $R = 3m$, what depth must the tank be filled to so that it holds $30m^3$? Use three iterations of the Newton-Raphson method to determine your answer. Determine the approximate relative error after each iteration.

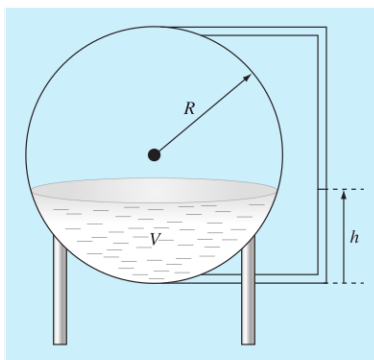


Figure 1: Water tank

5 Problem 5

The polynomial $f(x) = 0.0074x^4 - 0.284x^3 + 3.355x^2 - 12.183x + 5$ has a real root between 15 and 20. Apply the Newton-Raphson method to this function using an initial guess of $x_0 = 16.15$. Explain your results.

6 Problem 6

Use the secant method on the circle function $(x + 1)^2 + (y - 2)^2 = 16$ to find a positive real root. Set your initial guess to $x_i = 3$ and $x_{i-1} = 0.5$. Approach the solution from the first and fourth

quadrants. When solving for $f(x)$ in the fourth quadrant, be sure to take the negative value of the square root. Why does your solution diverge?

7 Problem 7

In control systems analysis, transfer functions are developed that mathematically relate the dynamics of a system's input to its output. A transfer function for a robotic positioning system is given by,

$$G(s) = \frac{C(s)}{N(s)} = \frac{s^3 + 12.5s^2 + 50.5s + 66}{s^4 + 19s^3 + 122s^2 + 296s + 192} \quad (5)$$

where $G(s)$ = system gain, $C(s)$ = system output, $N(s)$ = system input, and s = Laplace transform complex frequency. Use a numerical technique to find the roots of the numerator and denominator and factor these into the form,

$$G(s) = \frac{(s + a_1)(s + a_2)(s + a_3)}{(s + b_1)(s + b_2)(s + b_3)(s + b_4)} \quad (6)$$

where a_i and b_i are roots of the numerator and denominator respectively.

8 Problem 8

The general form for a three-dimensional stress field is given by,

$$\begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_{zz} \end{bmatrix} \quad (7)$$

where the diagonal terms represent tensile or compressive stresses and the off-diagonal terms represent shear stresses. A stress field (in MPa) is given by

$$\begin{bmatrix} 10 & 14 & 25 \\ 14 & 7 & 15 \\ 25 & 15 & 16 \end{bmatrix} \quad (8)$$

To solve for the principal stresses, it is necessary to construct the following matrix (again in MPa):

$$\begin{bmatrix} 10 - \sigma & 14 & 25 \\ 14 & 7 - \sigma & 15 \\ 25 & 15 & 16 - \sigma \end{bmatrix} \quad (9)$$

σ_1 , σ_2 and σ_3 can be solved from the equation,

$$\sigma^3 - I_1\sigma^2 + I_2\sigma - I_3 = 0 \quad (10)$$

where

$$I_1 = \sigma_{xx} + \sigma_{yy} + \sigma_{zz} \quad (11)$$

$$I_2 = \sigma_{xx}\sigma_{yy} + \sigma_{yy}\sigma_{zz} + \sigma_{zz}\sigma_{xx} - \sigma_{xy}^2 - \sigma_{yz}^2 - \sigma_{zx}^2 \quad (12)$$

$$I_3 = \sigma_{xx}\sigma_{yy}\sigma_{zz} + 2\sigma_{xy}\sigma_{yz}\sigma_{zx} - \sigma_{xx}\sigma_{yz}^2 - \sigma_{yy}\sigma_{zx}^2 - \sigma_{zz}\sigma_{xy}^2, \quad (13)$$

I_1, I_2, I_3 are the stress invariants. Find σ_1, σ_2 , and σ_3 using a root-finding technique.

9 Problem 9

In the thermos shown in Fig. 2, the innermost compartment is separated from the middle container by a vacuum. There is a final shell around the thermos. This final shell is separated from the middle layer by a thin layer of air. The outside of the final shell comes in contact with room air. Heat transfer from the inner compartment to the next layer q_1 is by radiation only (since the space is evacuated). Heat transfer between the middle layer and outside shell q_2 is by convection in a small space. Heat transfer from the outside shell to the air q_3 is by natural convection. The heat flux from each region of the thermos must be equal—that is, $q_1 = q_2 = q_3$. Find the temperatures T_1 and T_2 at steady state. T_0 is $450C$ and $T_3 = 25C$.

$$q_1 = 10^{-9} [(T_0 + 273)^4 - (T_1 + 273)^4] \quad (14)$$

$$q_2 = 4(T_1 - T_2) \quad (15)$$

$$q_3 = 1.3(T_2 - T_3)^{4/3} \quad (16)$$

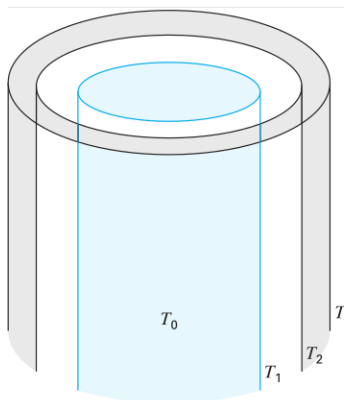


Figure 2: Thermos

10 Problem 10

Of the following three sets of linear equations, identify the set(s) that you could not solve using an iterative method such as Gauss-Seidel. Show using any number of iterations that is necessary that

Set One	Set Two	Set Three
$9x + 3y + z = 13$	$x + y + 6z = 8$	$-3x + 4y + 5z = 6$
$-6x + 8z = 2$	$x + 5y - z = 5$	$-2x + 2y - 4z = -3$
$2x + 5y - z = 6$	$4x + 2y - 2z = 4$	$2y - z = 1$

Table 1: Systems of Linear Equations: Set One, Set Two, and Set Three

your solution does not converge. Clearly state your convergence criteria (how you know it is not converging).

11 Problem 11

Consider the three mass-four spring system in Fig. 3. Determining the equations of motion from $\sum F_x = ma$, for each mass using its free-body diagram results in the following differential equations:

$$\begin{aligned}
 \ddot{x}_1 + \frac{k_1 + k_2}{m_1}x_1 - \frac{k_2}{m_1}x_2 &= 0, \\
 \ddot{x}_2 - \frac{k_2}{m_2}x_1 + \frac{k_2 + k_3}{m_2}x_2 - \frac{k_3}{m_2}x_3 &= 0, \\
 \ddot{x}_3 - \frac{k_3}{m_3}x_2 + \frac{k_3 + k_4}{m_3}x_3 &= 0.
 \end{aligned} \tag{17}$$

where $k_1 = k_4 = 10N/m$, $k_2 = k_3 = 30N/m$, and $m_1 = m_2 = m_3 = 2kg$. Write the three equations in matrix form:

$$0 = [\text{Acceleration vector}] + [k/m \text{ matrix}][\text{displacement vector } x]$$

At a specific time where $x_1 = 0.05m$, $x_2 = 0.04m$, and $x_3 = 0.03m$, this forms a tridiagonal matrix. Solve for the acceleration of each mass.

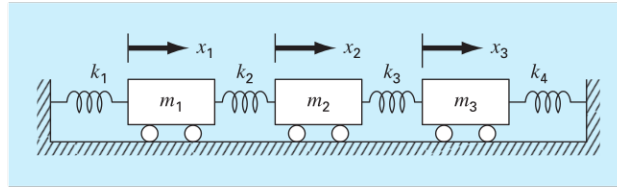


Figure 3: Spring mass system

12 Problem 12

The steady-state distribution of temperature on a heated plate can be modeled by the Laplace equation,

$$0 = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \quad (18)$$

If the plate is represented by a series of nodes (Fig. 4), centered finite-divided differences can be substituted for the second derivatives, which results in a system of linear algebraic equations. Use the Gauss-Seidel method to solve for the temperatures of the nodes in Fig. 4.

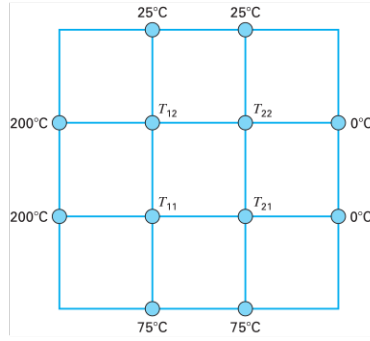


Figure 4: Temperature grid.